

The Hand

Department of Anatomy

Musculoskeletal System

The Upper Limb (5)

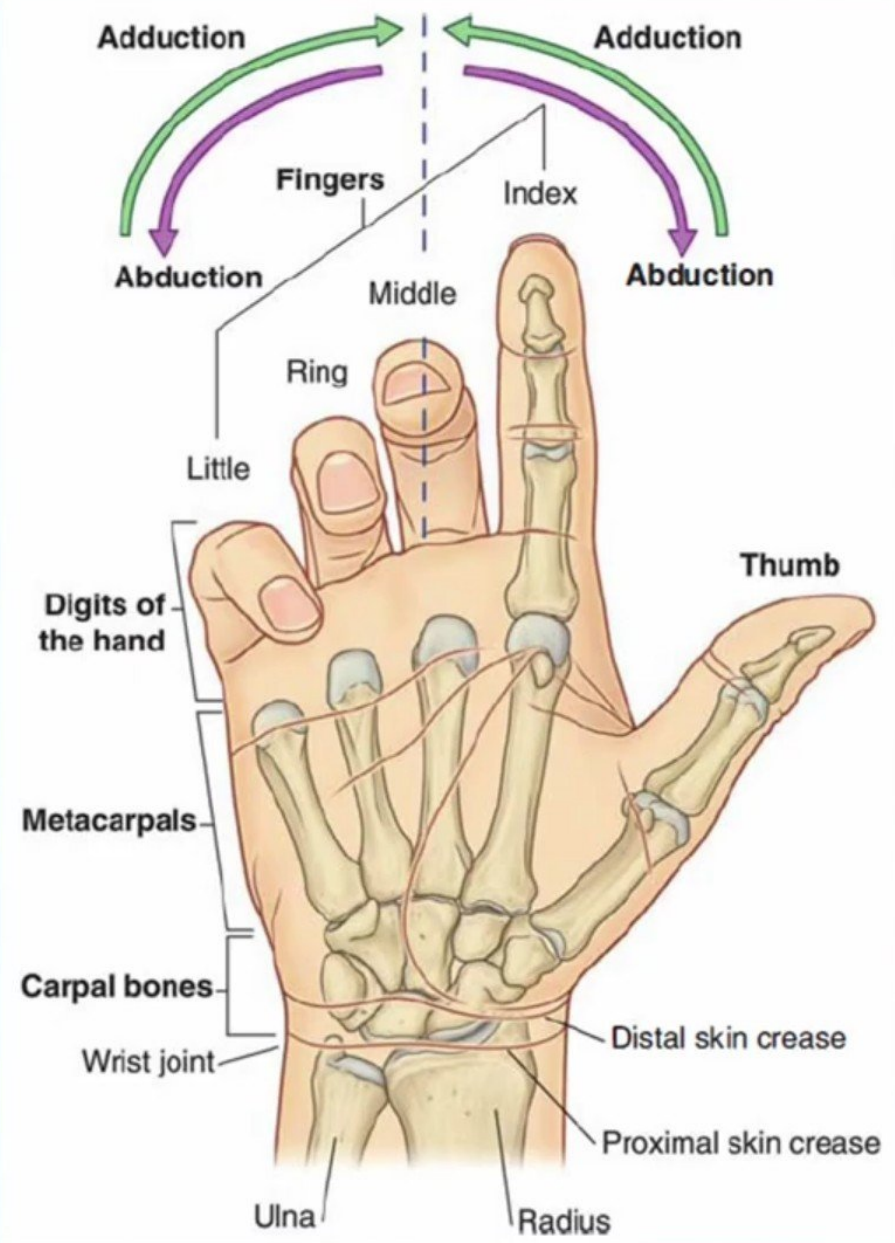
First Part

Dr. Osman Ahmed Osman

Lecture points

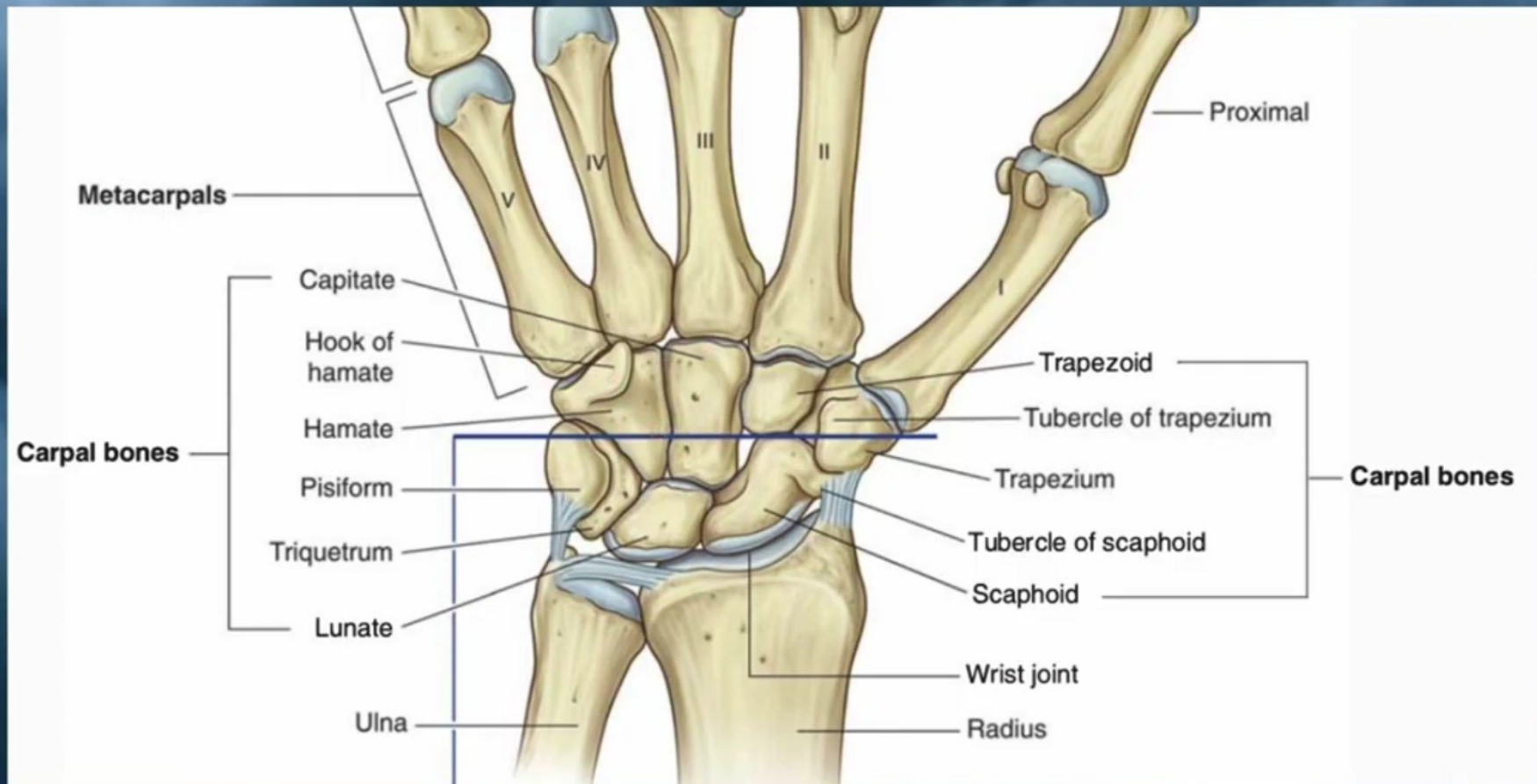
- Revision bone of the hand.
 - Revision movement of the hand (Movements of the thumb).
 - The Palmar aponeurosis
 - Flexor retinaculum.
 - The carpal tunnel.
 - Muscles of the hand
 - Extrinsic Hand Muscles.
 - Intrinsic Hand Muscles.
 - Thenar muscles.
 - Hypothenar muscles.
 - Other.
- Blood supply
 - Arterial by (superficial and deep palmar arches).
 - The Dorsal venous arch of the hand.
 - Lymph Drainage of the Palm
 - Innervation of the hand.
 - Fascial Palmer Spaces

Revision bone of the hand

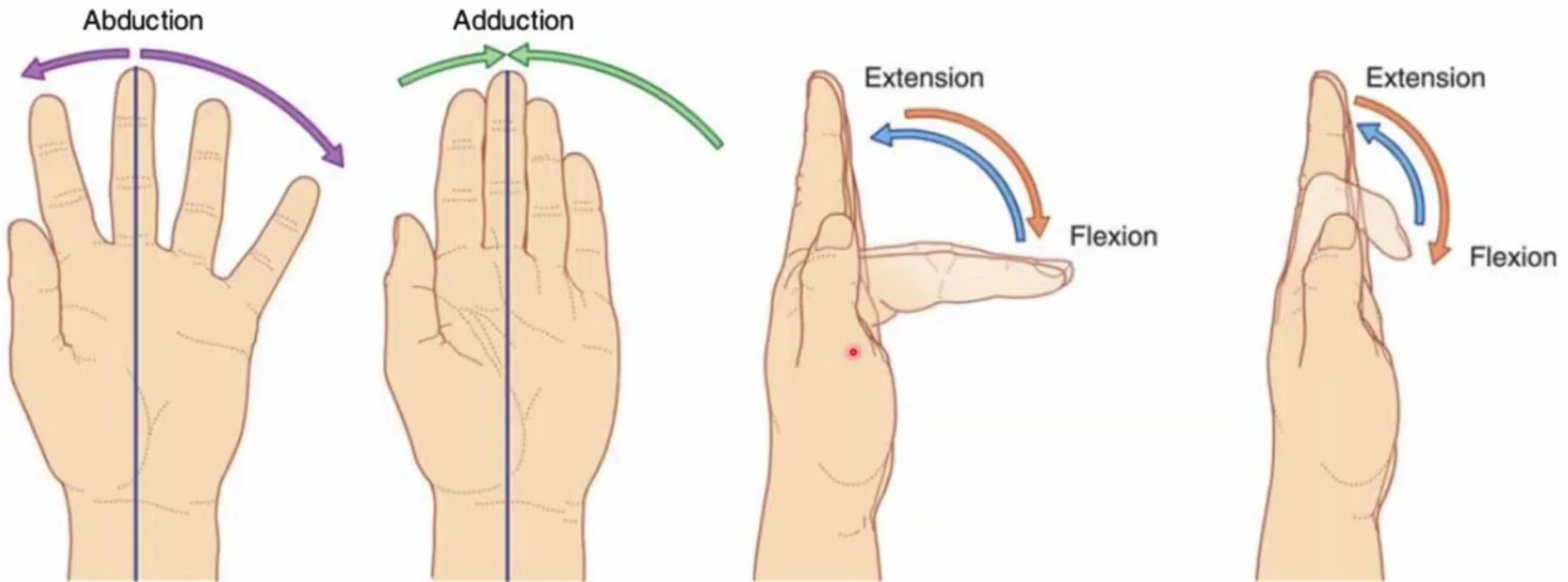


Carpal Bones

Scapoid
Lunate
Triquetrium
Pisiform
Trapeziod
Trapezium
Capitate
Hamate

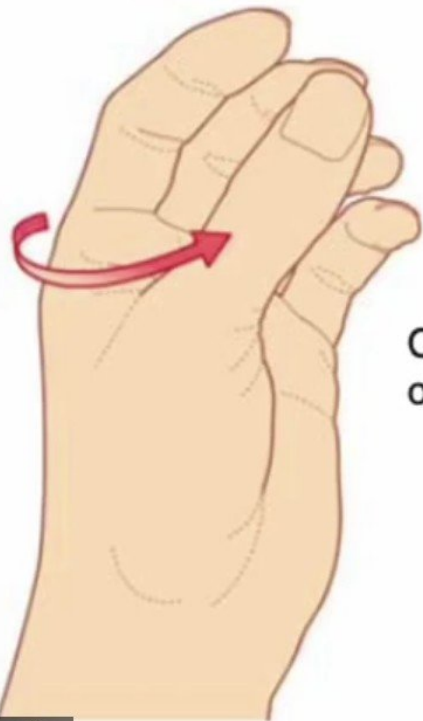


Revision movement of the hand

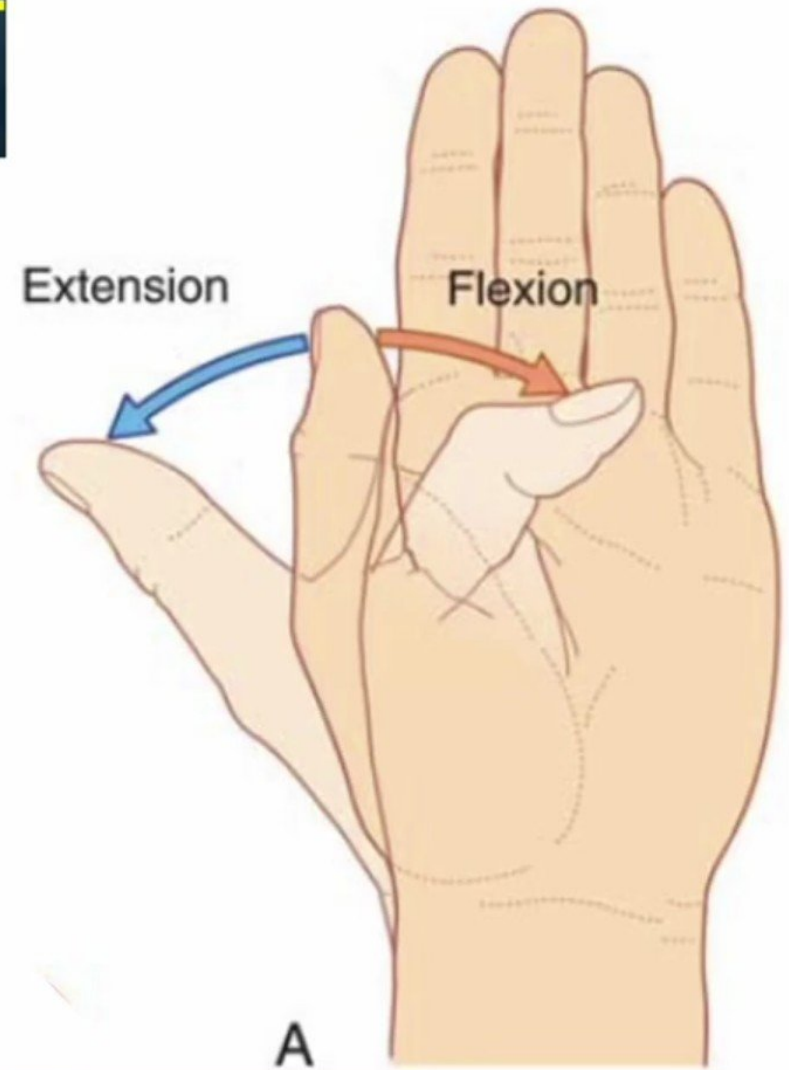
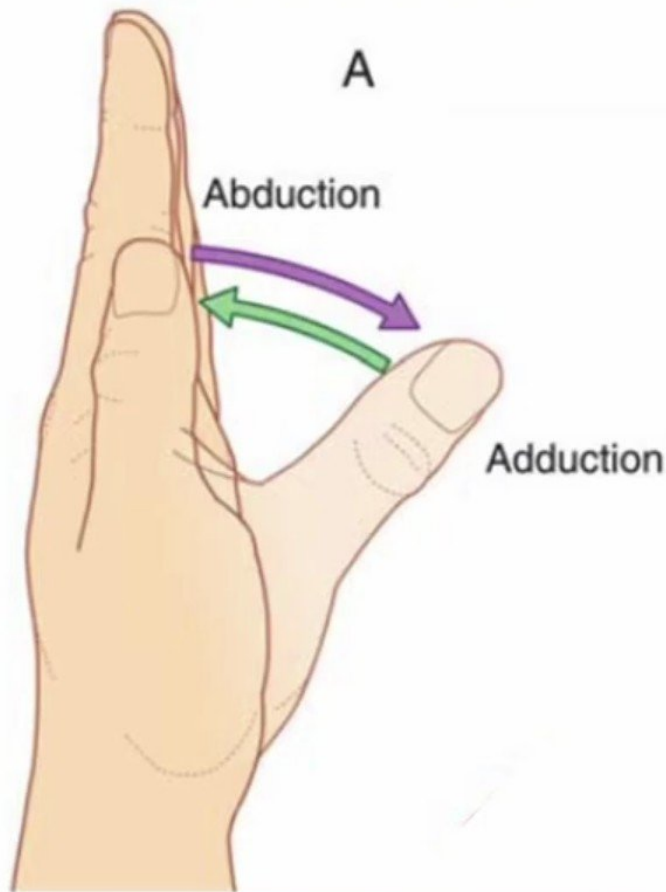


Revision movement of the hand

Movements of the thumb



Opposition of thumb



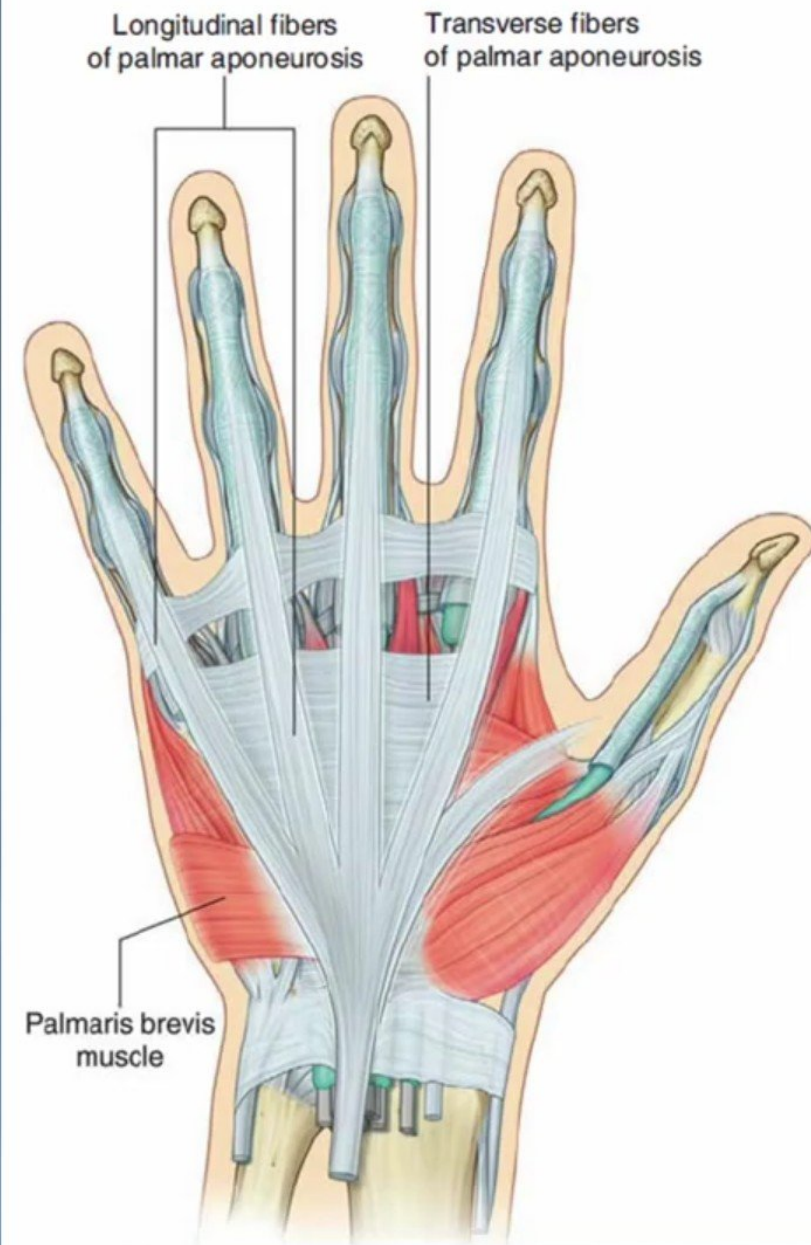
The Palmar aponeurosis

➤ It is **deep fascia** in the **central region of the palm** is reinforced by:

➤ The superficial layer of longitudinal fibres continuous with the tendon of the palmaris longus muscle.

✓ The longitudinal fibres are usually **present** even **when palmaris is absent**.

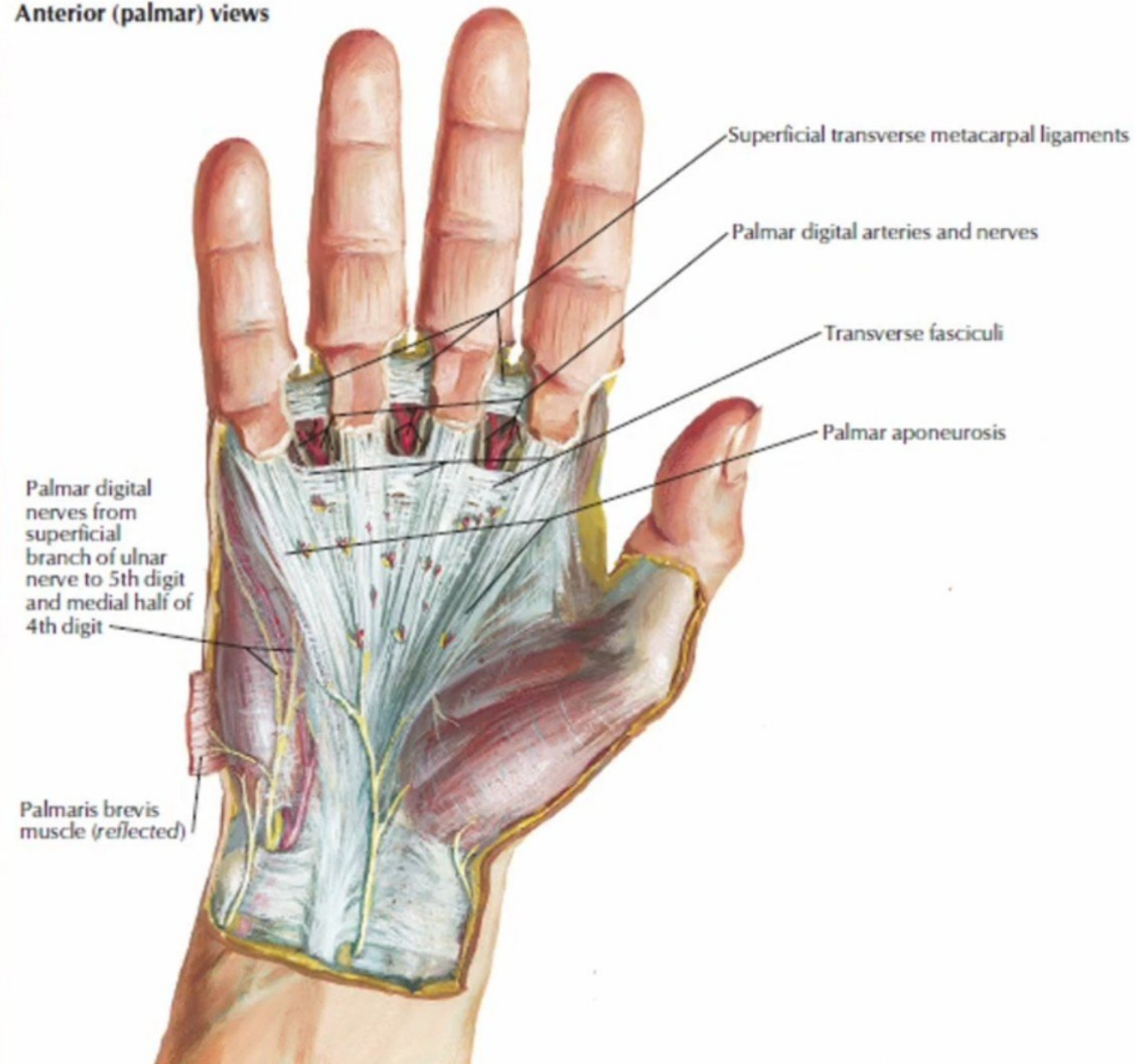
➤ The deeper transverse fibres.



Palmar aponeurosis

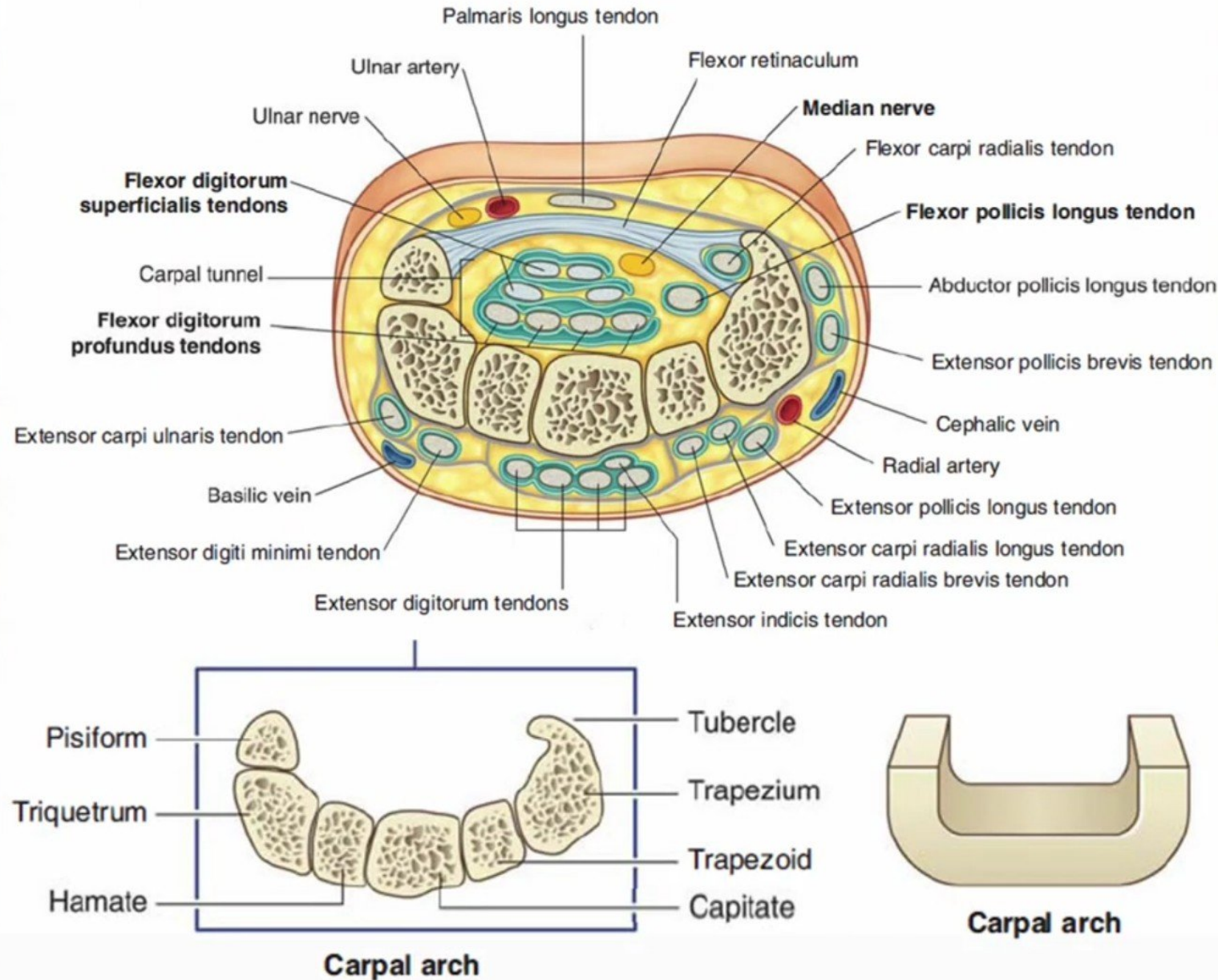
- It is **continuous proximally** with **the flexor retinaculum** and **on either side** with **thinner fascia** covering **the thenar** and **hypothenar muscles**.
- It **widens distally** in the hand and **divides into four strips**, one for each finger.
- The main part of each digital slip divides into two diverging bands that are inserted into:
 - **The deep transverse metacarpal ligament (natatory ligament)..**
 - **The bases of the proximal phalanx.**
 - **The fibrous flexor sheath.**

Anterior (palmar) views



Flexor retinaculum

- It is a strong **fibrous band**,
- It lies across the front of the carpus at the **proximal part of the hand**.
- it is thold the long flexor tendons in position at the wrist.
- Its proximal limit lies at the level of the distal, dominant skin crease on the front of the wrist.
- It is attached to **the hook of the hamate** and **the pisiform medially** and to the **tubercle of the scaphoid** and the **ridge of the trapezium laterally**.

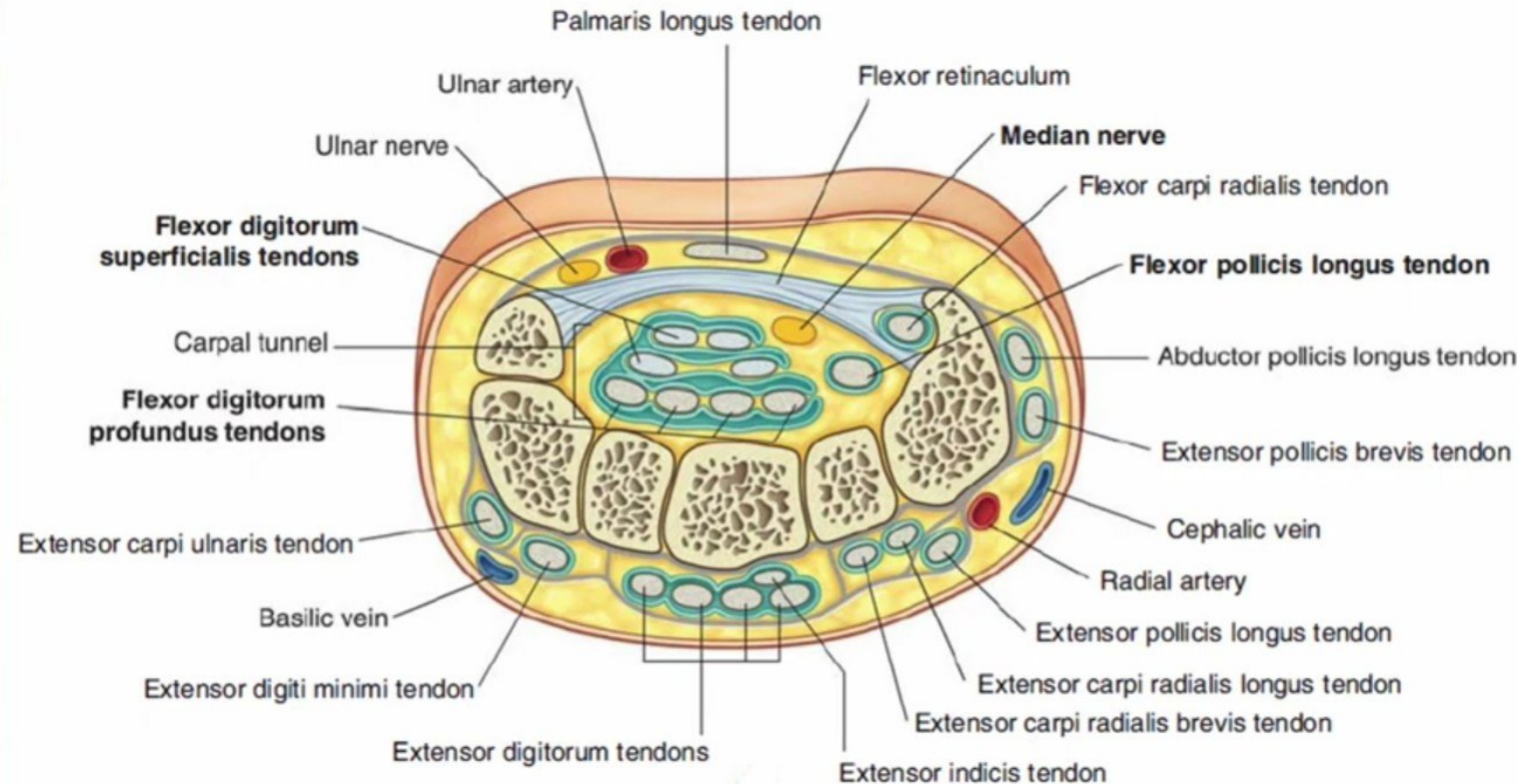


The carpal tunnel

- It lies between **the flexor retinaculum and the carpal bones**.
- **The structures pass below the flexor retinaculum is:**

- 1) The median nerve.
 - 2) The all long flexor tendons of the fingers and thumb.
- ✓ The tendon of flexor digitorum superficialis.
 - ✓ The tendon of flexor digitorum profundus.
 - ✓ The tendon of flexor pollicis longus.

➤ **The tendon of flexor carpi radialis**, runs in the groove in this sub-compartment of the carpal tunnel.



- **The structures pass above the flexor retinaculum is:**

- **Ulnar artery**
- **Ulnar nerve.**
- **Palmaris longus tendon.**
- **Superficial palmar cutaneous branch of the median nerve.**
- **Superficial palmar cutaneous branch of the ulnar nerve**

Extrinsic Hand Muscles

- Extensor Indicis.
- Extensor Pollicis Longus.
- Extensor Pollicis Brevis.
- Abductor Pollicis Longus.
- Flexor Pollicis Longus.
- Extensor Digiti Minimi
- Flexor Digitorum Superficialis
- Flexor Digitorum Profundus

Intrinsic Hand Muscles

- **Thenar Eminance**
 - ✓ Abductor Pollicis Brevis
 - ✓ Flexor Pollicis Brevis
 - ✓ Opponens Pollicis .
- **Hypothenar Eminence.**
 - ✓ Abductor Digiti Minimi.
 - ✓ Opponen Digiti Minimi.
 - ✓ Flexor Digiti Minimi
- **Other Intrinsic Hand Muscles:**
 - ✓ Adductor Pollicis.
 - ✓ Palmar Interossei.
 - ✓ Dorsal Interossei.
 - ✓ Lumbricales

Extensor Indicis

➤ Origin:

- ✓ Dorsal surface lower ½ body of ulna.
- ✓ Interosseus membrane.

➤ Insertion:

- ✓ Ulnar side of index finger's EDC tendon.

➤ Nerve supply:

- ✓ Radial (posterior interosseous) nerve.

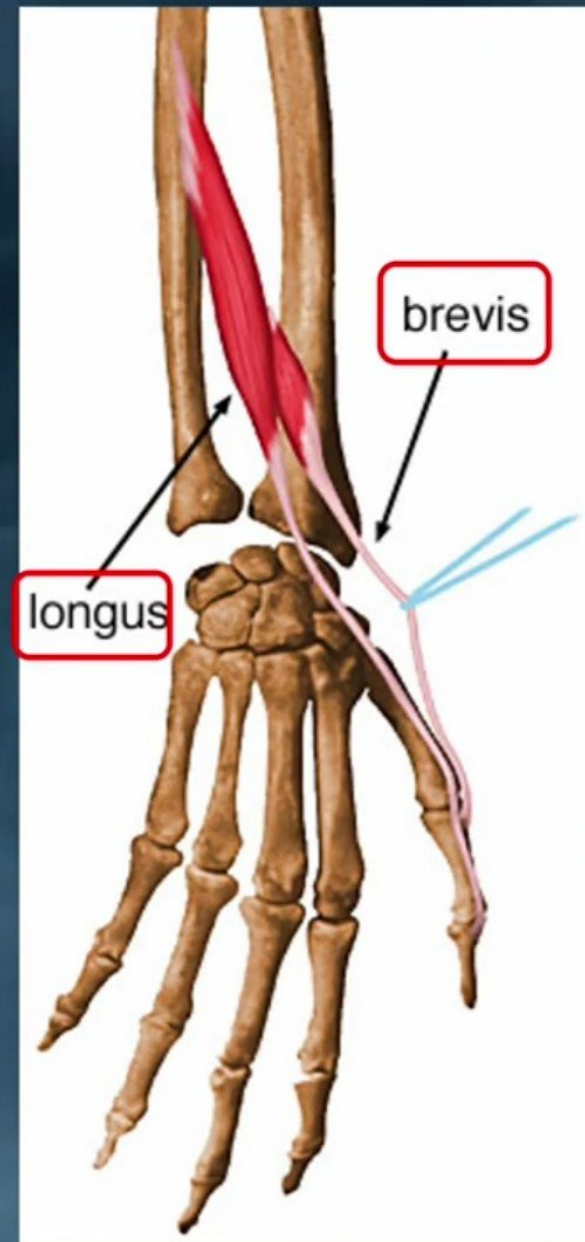
➤ Action:

- ✓ Extends metacarpophalangeal and interphalangeal joint of index finger.



Extensor Pollicis Longus

- **Origin:**
 - Posterior 1/3 ulna
 - Interosseus membrane
- **Insertion:**
 - Posterior surface of base of thumb distal phalanx
- **Nerve supply:**
 - Radial (posterior interosseus)
- **Action:**
 - Extends distal phalanx of thumb.
 - Extends Carpal metacarpal , metcarophalangeal and interphalangeal joint of digit thumb.
 - The tendon is seen and felt on the ulnar side of the snuffbox.



Extensor Pollicis Brevis

➤ Origin:

- ✓ Dorsal 2/3 of radius

➤ Insertion:

- ✓ Dorsal surface of base of proximal 1st phalanx

➤ Nerve supply:

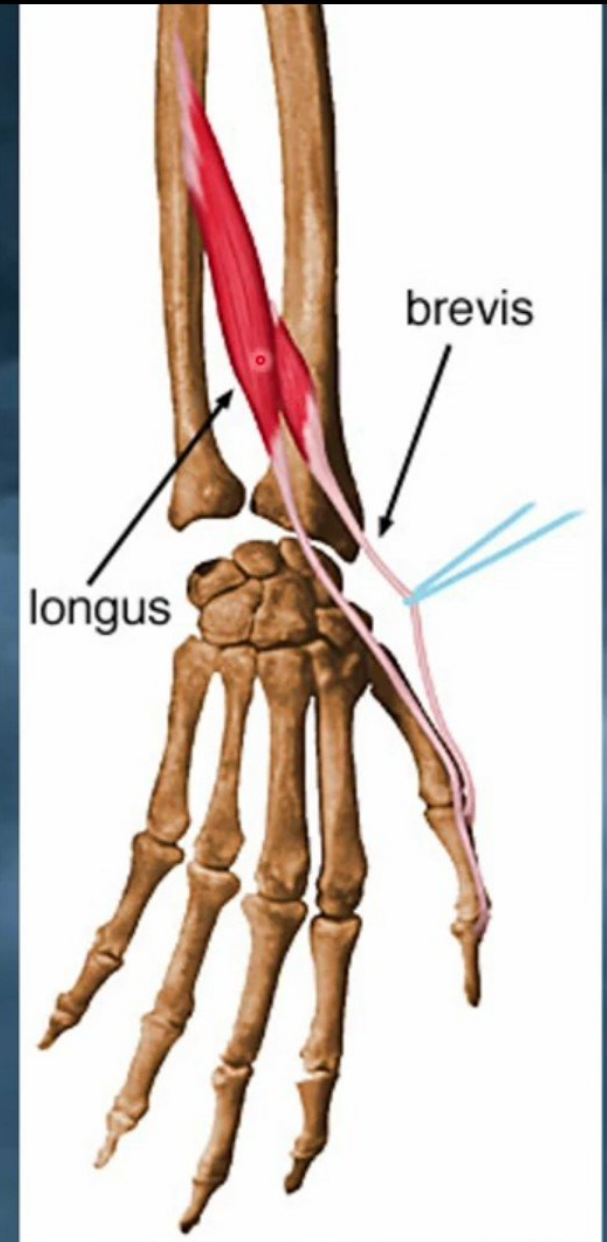
- ✓ Radial (posterior interosseus)

➤ Action:

- ✓ It extends the carpometacarpal and metacarpophalangeal joints of the thumb.

➤ Test.

- ✓ When the thumb is extended at the metacarpophalangeal joint against resistance, the tendon is seen and felt at the radial side of the snuffbox on the ulnar side of the adjacent abductor pollicis longus tendon.



Abductor Pollicis Longus

➤ Origin:

- ✓ Posterior distal 2/3 of ulna
- ✓ Posterior middle 1/3 of radius
- ✓ Interosseus membrane.

➤ Insertion:

- ✓ Radial side of base of 1st metacarpal.

➤ Nerve supply:

- ✓ Radial (posterior interosseous nerve).

➤ Action:

- ✓ It extends the thumb at the carpometacarpal joint, displacing it laterally in the plane of the palm.
- ✓ It can assist in abducting and flexing the wrist, producing a 'trick' flexion when other flexors are paralysed.

➤ Test:

- ✓ when thumb is extended at the carpometacarpal joint against resistance, the tendon is felt at the radial side of the snuffbox and on the radial side of the adjacent extensor pollicis brevis tendon.



Flexor Pollicis Longus

➤ Origin:

- ✓ Anterior middle ½ of radius
- ✓ Interosseus membrane

➤ Insertion:

- ✓ Palmar surface of base of distal 1st phalanx

➤ Nerve supply:

- ✓ Median (anterior interosseous)

➤ Action:

- ✓ It is the only flexor of the interphalangeal joint of the thumb, and also flexes the metacarpophalangeal and carpometacarpal joints of the thumb and the wrist joint.

➤ Test:

- ✓ With the proximal phalanx of the thumb held steady, the distal phalanx is flexed against resistance.



Extensor Digiti Minimi

➤ Origin:

- ✓ Lateral epicondyle of humerus

➤ Insertion:

- ✓ Extensor expansion of the little finger.

➤ Nerve supply:

- ✓ Radial (posterior interosseous)

➤ Action:

- ✓ It extends the metacarpophalangea and interphalangeal joint of the little finger.
- ✓ It assists extensor digitorum in extension of the little finger and wrist joint.



Flexor Digitorum Superficialis

➤ Origin:

- ✓ Medial epicondyle of humerus
- ✓ Coronoid process
- ✓ Middle ½ anterior radius

➤ Insertion:

- ✓ Four tendons separating into two parts that insert into sides of bases of middle 2-5 phalanxes

➤ Nerve supply:

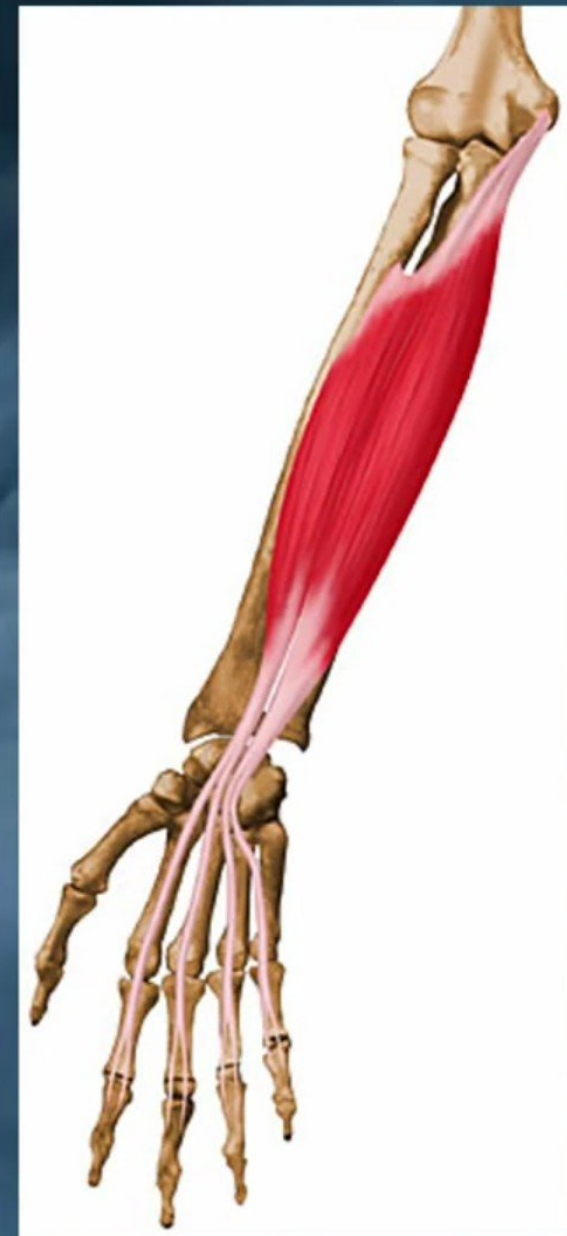
- ✓ Median

➤ Action:

- ✓ It is a flexor of the proximal interphalangeal joints, and secondarily of the metacarpophalangeal and wrist joints.
- ✓ It also assists in flexion of the elbow and wrist.

➤ Test:

- ✓ The fingers are flexed at the proximal interphalangeal joints against resistance applied to the middle phalanges, while the distal interphalangeal joints are kept extended.



Flexor Digitorum Profundus

➤ Origin:

- ✓ Anteromedial surface of ulna
- ✓ Interosseus membrane

➤ Insertion:

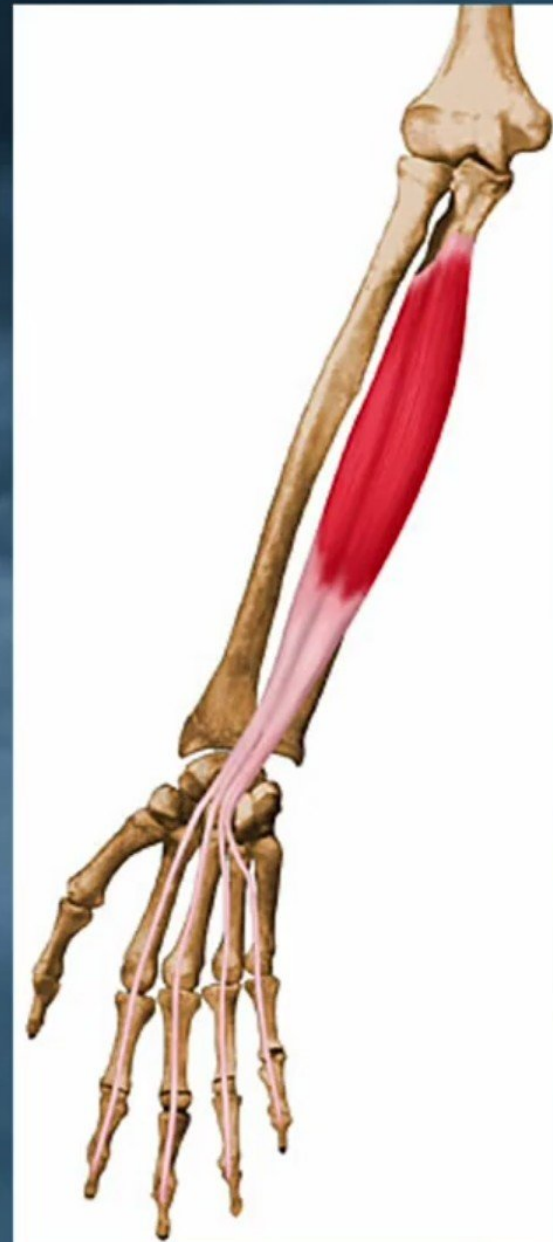
- ✓ Four tendons inserting into distal phalanxes of digits 2-5

➤ Nerve supply:

- ✓ Media 2-3 digits
- ✓ Ulna 4-5 digits

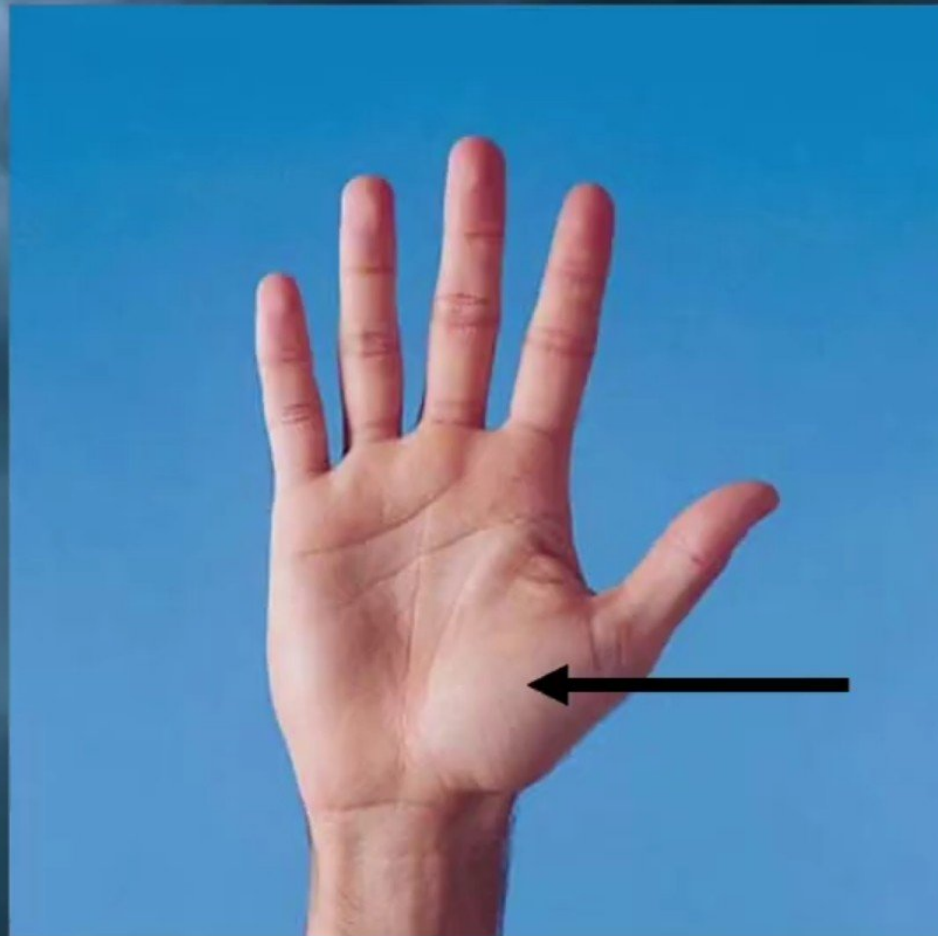
➤ Action:

- ✓ It flexes the terminal interphalangeal joints.
- ✓ It is the great gripping muscle.



Intrinsic Hand Muscles

Thenar Eminance



Abductor Pollicis Brevis

➤ Origin:

- ✓ Scaphoid tuberosity
- ✓ Trapezium ridge
- ✓ Transverse carpal ligament

➤ Insertion:

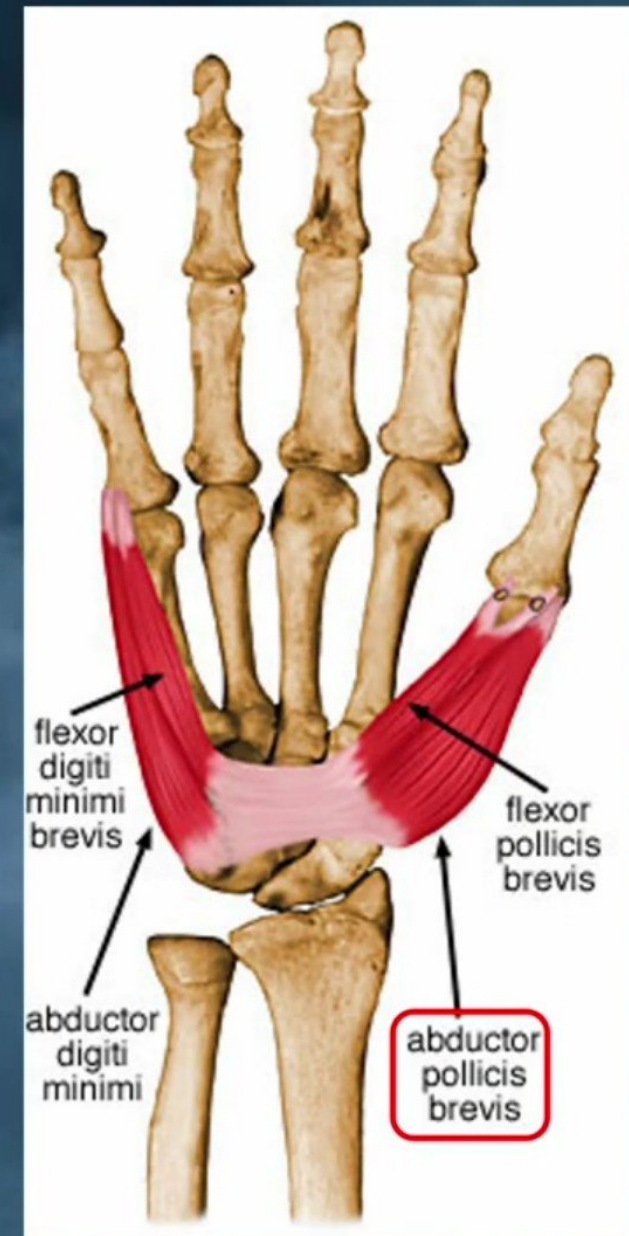
- ✓ Lateral base of proximal 1st phalanx

➤ Nerve supply:

- ✓ Median

➤ Action:

- ✓ Abduction of thumb in carpometacarpal and metacarpophalangeal joint.



Flexor Pollicis Brevis

➤ Origin:

- ✓ Superficial head – trapezium
- ✓ Deep head – trapezoid, capitate and palmar ligaments of distal carpal bones

➤ Insertion:

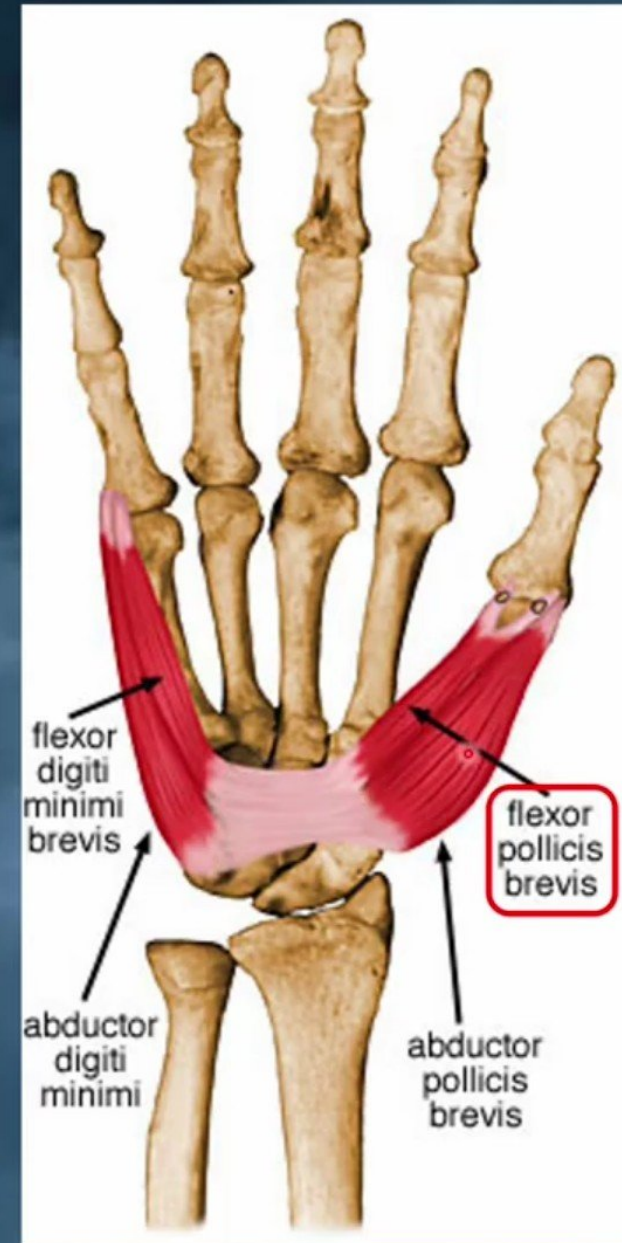
- ✓ Base of proximal 1st phalanx on radial side
- ✓ Extensor expansion

➤ Nerve supply:

- ✓ Superficial – median
- ✓ Deep – Ulnar

➤ Action:

- ✓ Flexion of thumb in carpometacarpal and metacarpophalangeal joint.



Opponens Pollicis

➤ Origin:

- ✓ Trapezium
- ✓ Transverse Carpal Ligament

➤ Insertion:

- ✓ Radial side of 1st metacarpal shaft

➤ Nerve supply:

- ✓ Median

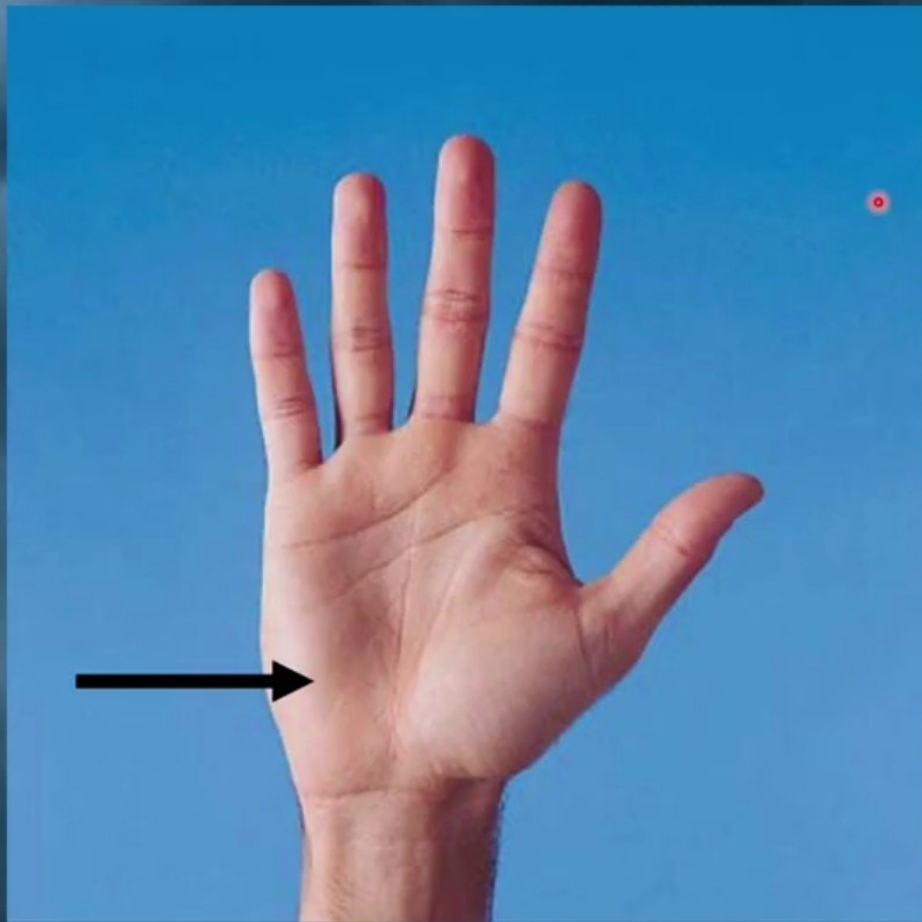
➤ Action:

- ✓ Opposition action on metacarpophalangeal joint by pulls thumb medially and forward across palm.



Intrinsic Hand Muscles

Hypothenar Eminence



Abductor Digiti Minimi

➤ Origin:

✓ Pisiform

➤ Insertion:

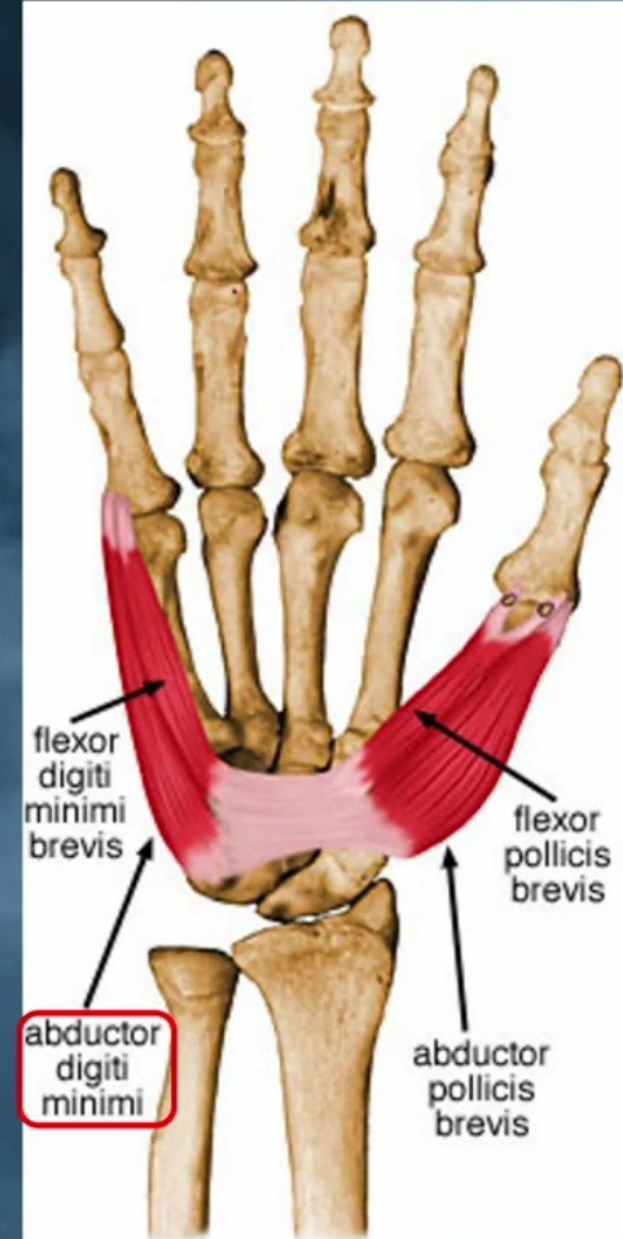
✓ Ulnar side base of 5th proximal phalanx

➤ Nerve supply:

✓ Ulnar

➤ Action:

✓ Abduction of the little finger in metacarpophalangeal joint.



Flexor Digiti Minimi

➤ Origin:

- ✓ Hamate bone
- ✓ Transverse carpal ligament

➤ Insertion:

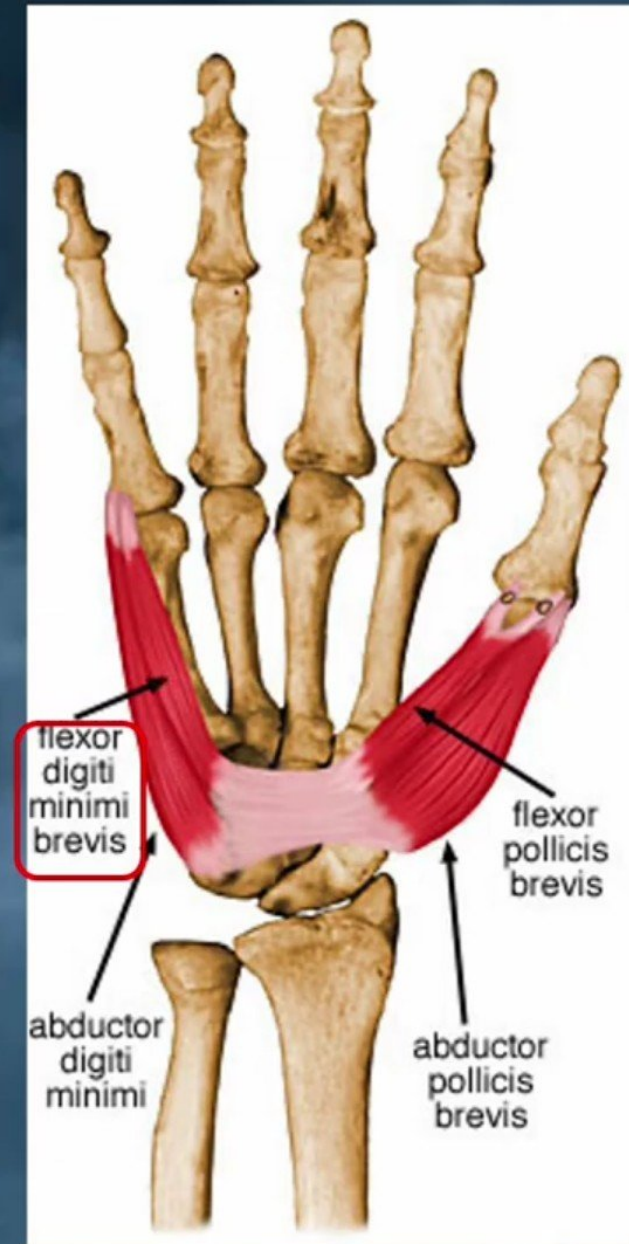
- ✓ Ulnar side of proximal 5th phalanx

➤ Nerve supply:

- ✓ Ulnar

➤ Action:

- ✓ Flexion of the little finger in metacarpophalangeal joint.



Opponen Digiti Minimi

➤ Origin:

- ✓ Hook of hamate
- ✓ Transverse carpal ligament

➤ Insertion:

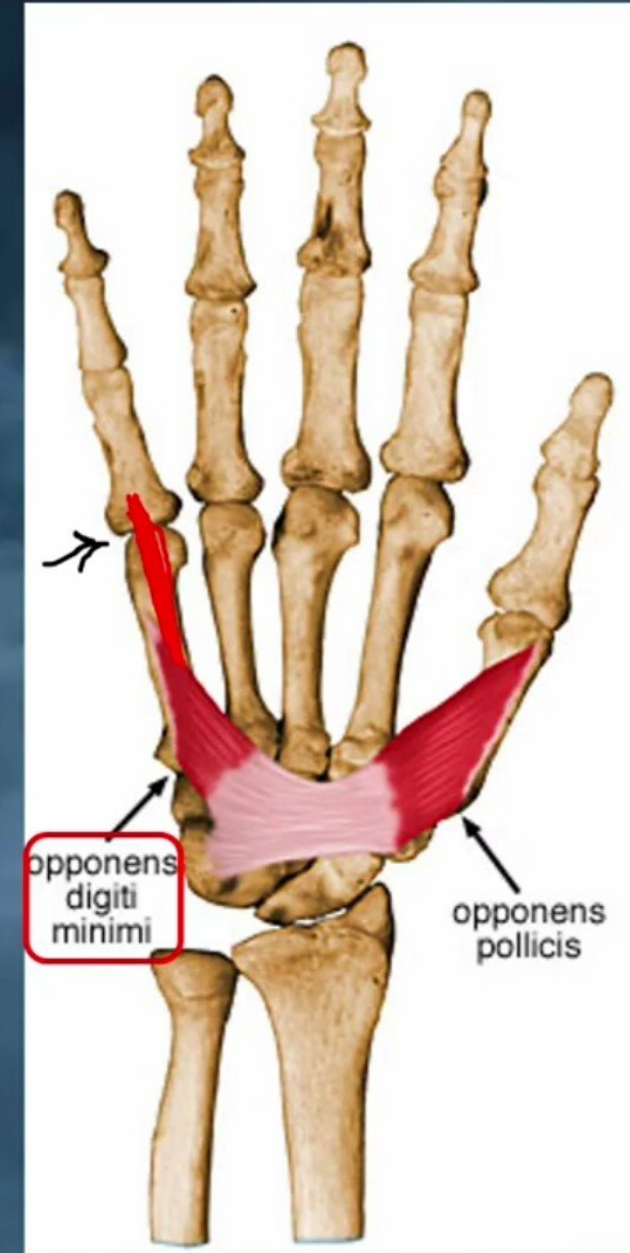
- ✓ Ulnar border of entire 5th metacarpal bone

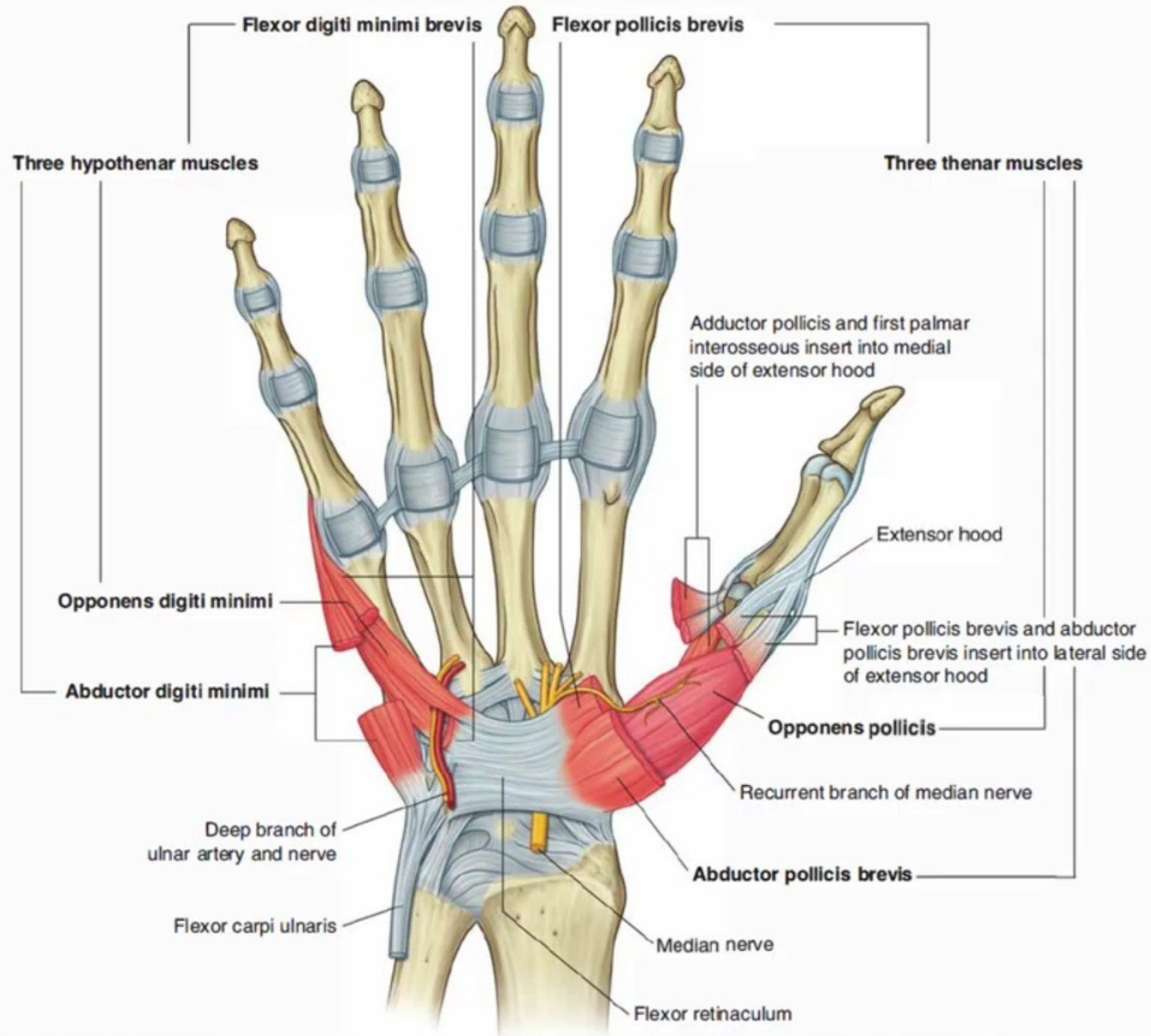
➤ Nerve supply:

- ✓ Ulnar

➤ Action:

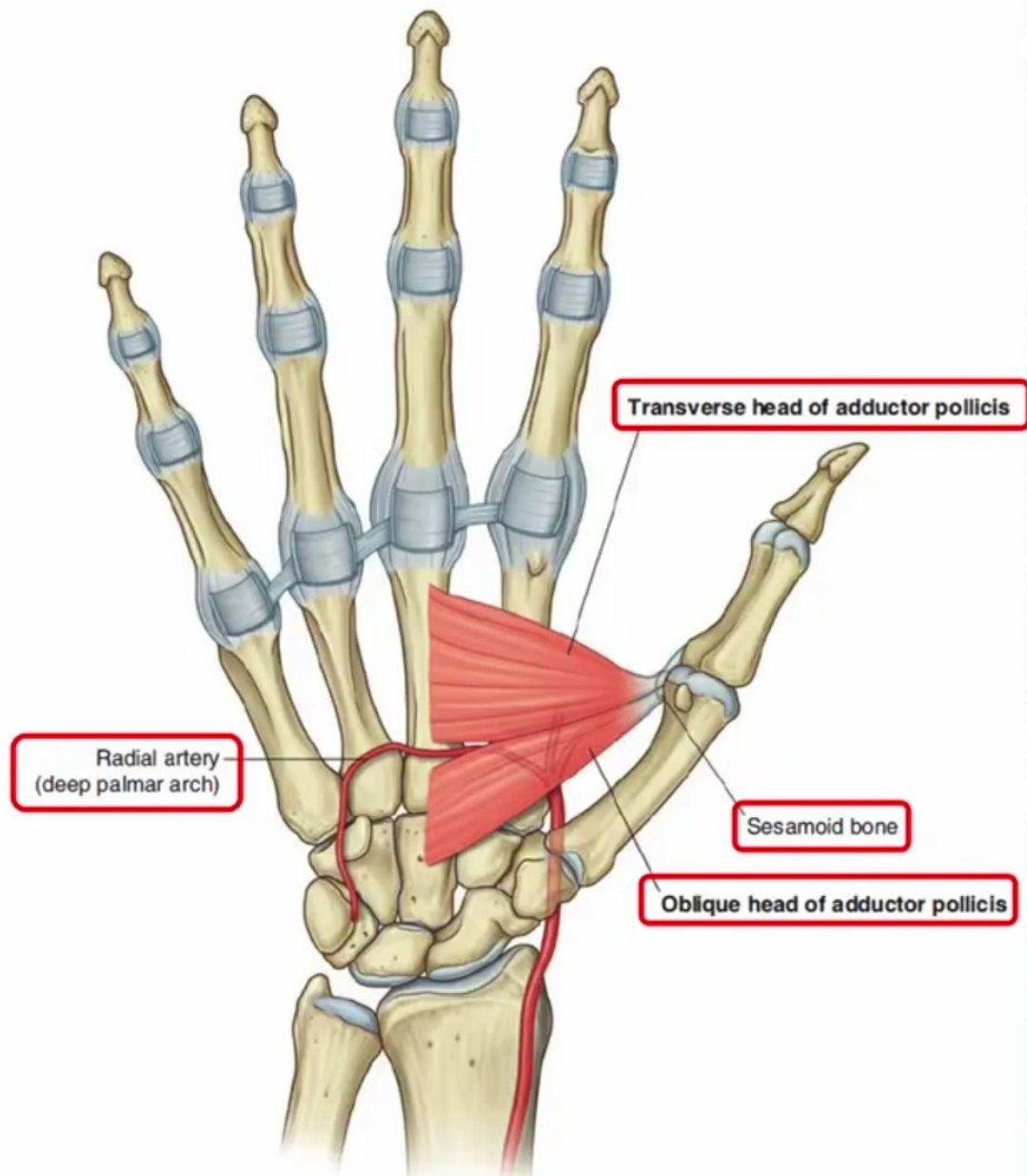
- ✓ Flexion & rotation of the little finger in metacarpophalangeal joint.





Other Intrinsic Hand Muscles

Adductor Pollicis



Adductor Pollicis

➤ Origin:

✓ Oblique Head

- Capitate bone
- Bases of 2-3 metacarpals

✓ Transverse Head

- Proximal 2/3 of palmar surface of 3rd metacarpal

➤ Insertion:

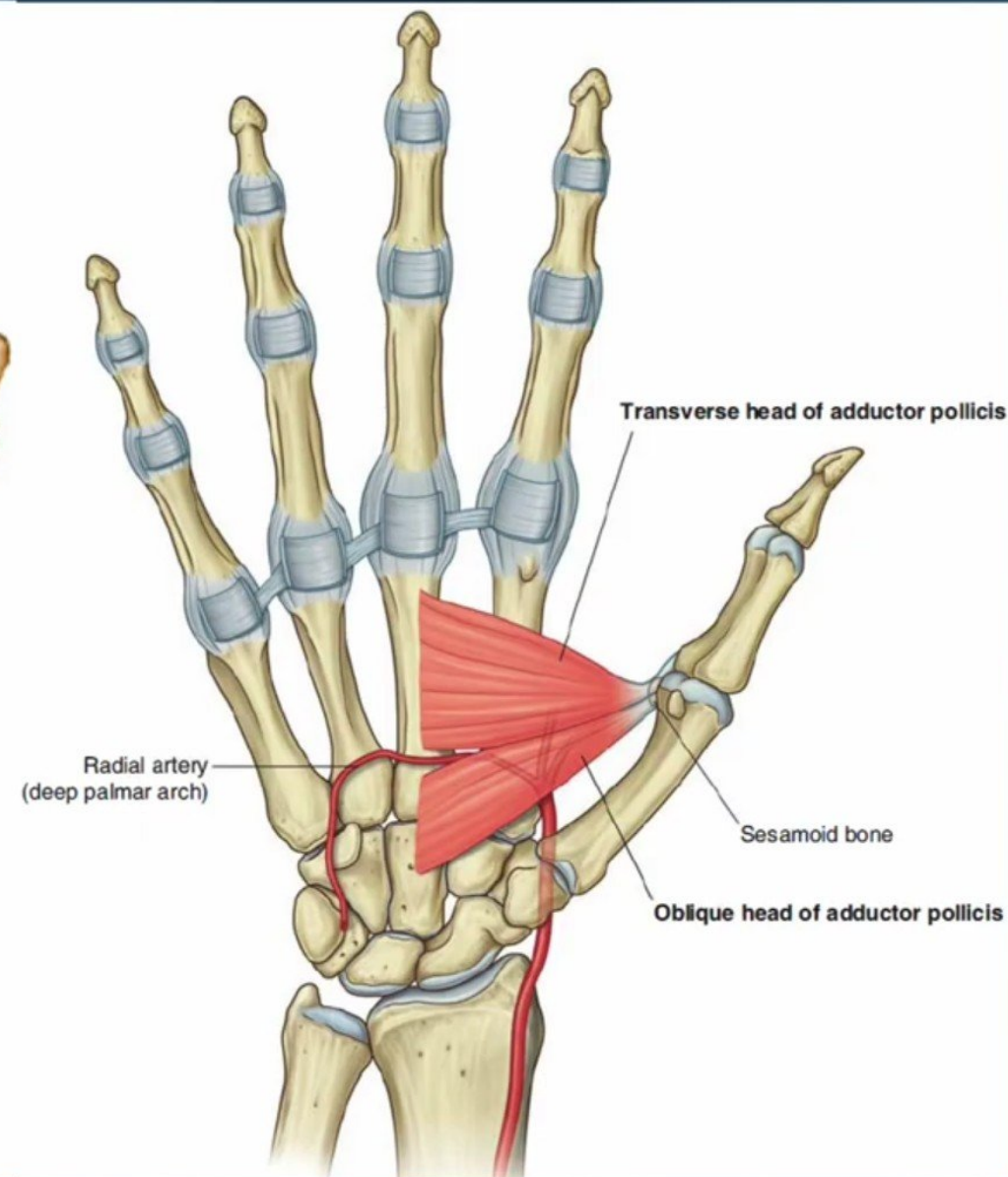
- ✓ Ulnar side of base of 1st proximal phalanx

➤ Nerve supply:

- ✓ Ulnar

➤ Action:

- ✓ Adduct of the thumb in carpometacarpal joint.



Palmar Interossei

➤ Origin:

- ✓ 1st – ulnar side base of 1st metacarpal bone.
- ✓ 2nd – ulnar side of 2nd metacarpal bone.
- ✓ 3rd – radial side of 4th metacarpal bone.
- ✓ 4th – radial side of 5th metacarpal bone.

➤ Insertion:

- ✓ Extensor expansion of 2,4 and 5th digits

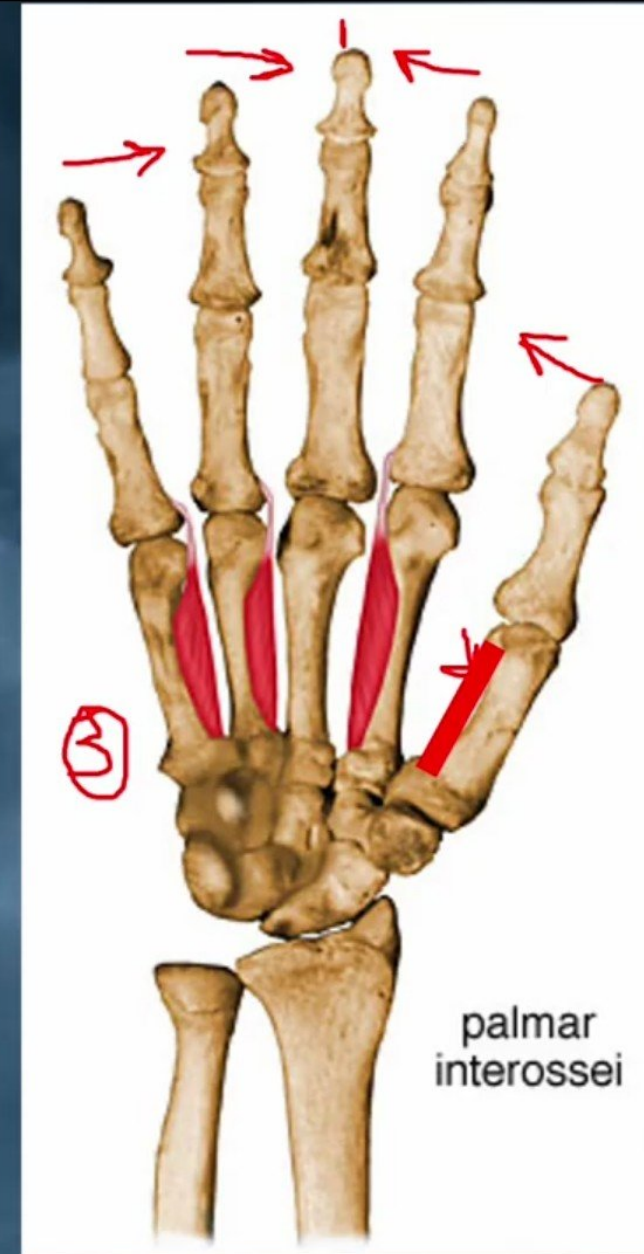
➤ Nerve supply:

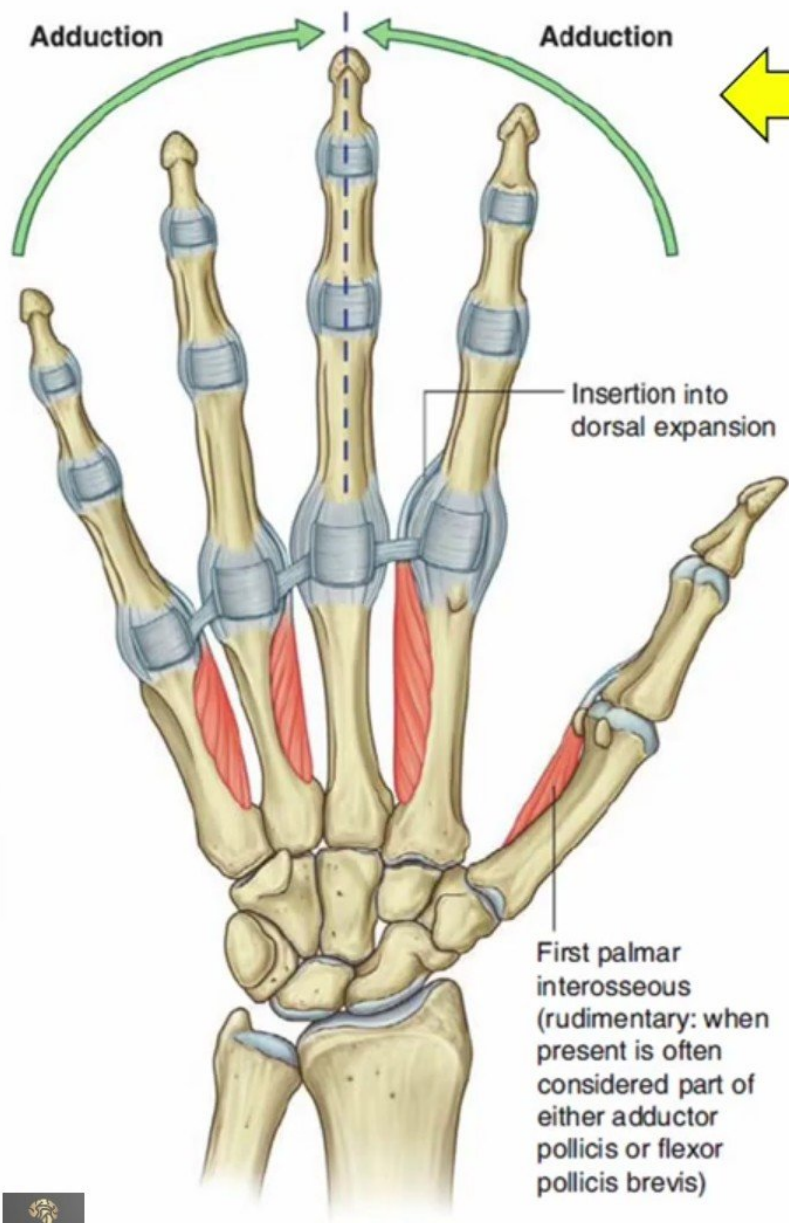
- ✓ Ulnar nerve.

➤ Action:

- ✓ Adduct fingers toward center of third finger.

- The **thumb** show **no palmar interosseous** as it already possesses its **own powerful adductor pollicis muscle**.



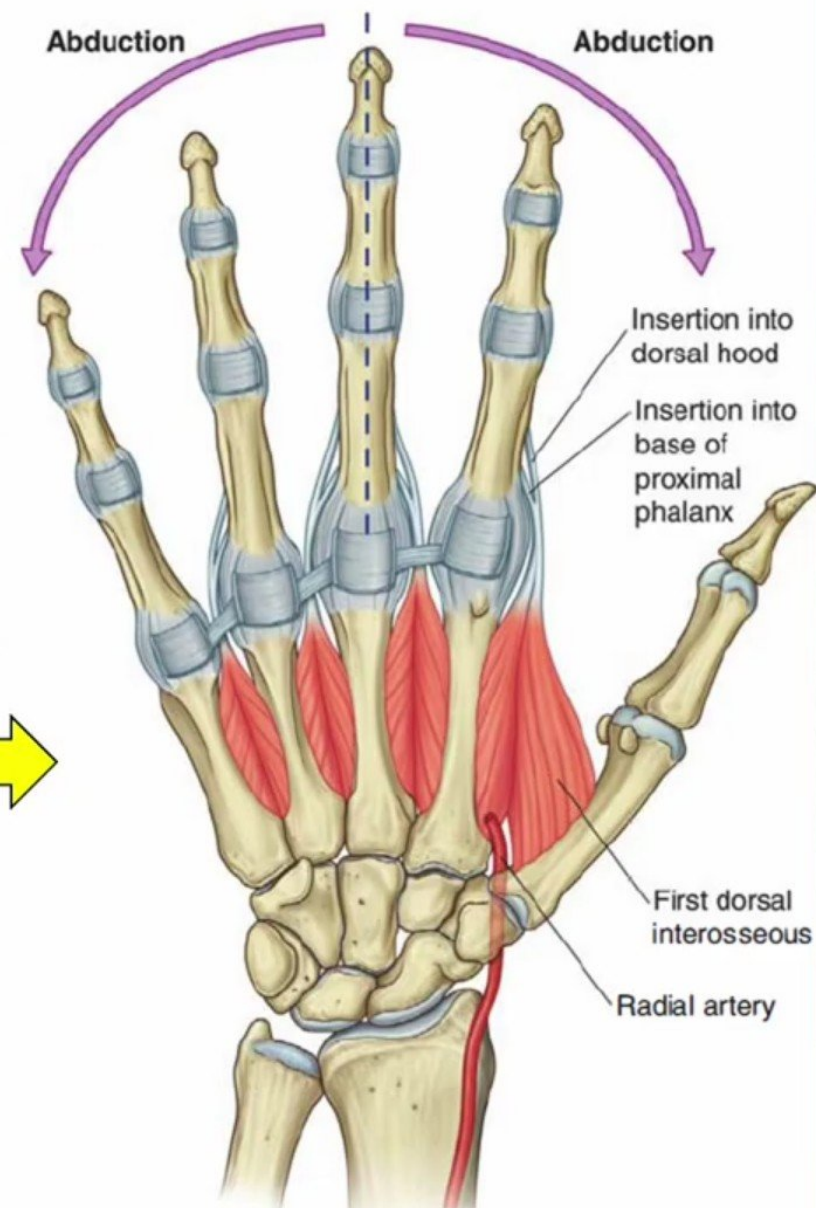
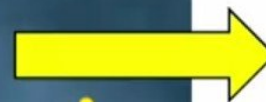


Palmar Interossei

PAD

Dorsal Interossei

DAB



Dorsal Interossei

➤ Origin:

- 1st lateral head – ulnar side of 1st metacarpal bone
- 1st medial head – radial side of 2nd metacarpal bone
- 2nd, 3rd, 4th space between metacarpal bones

➤ Insertion:

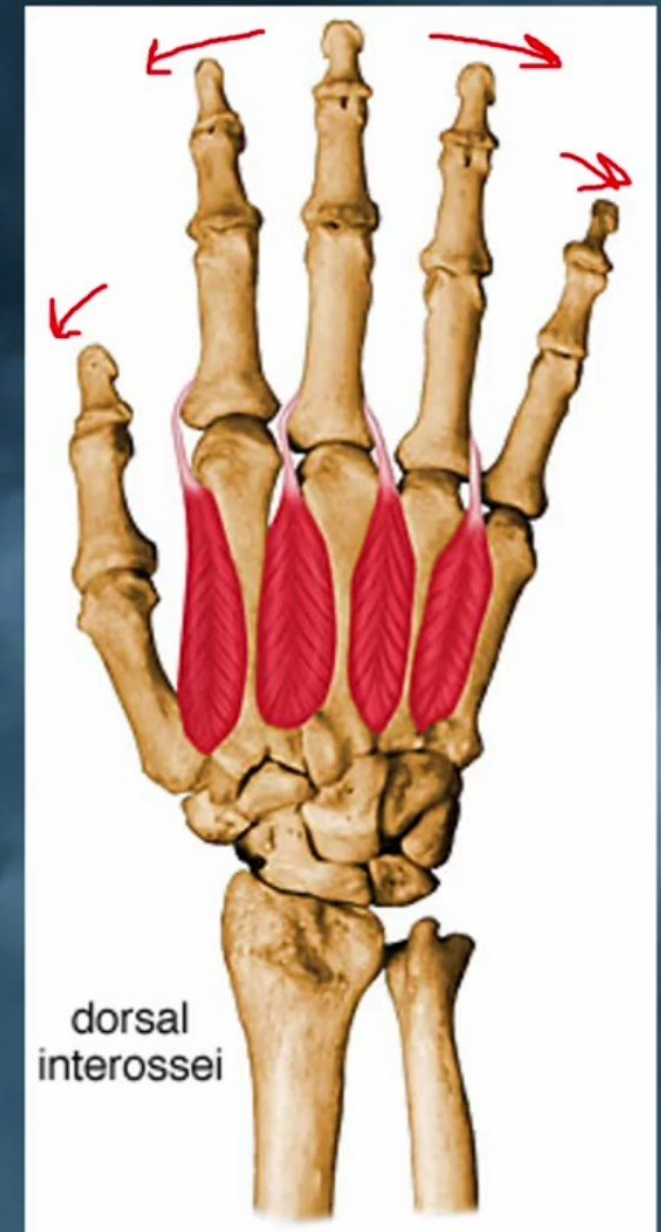
- 1st – radial side 2nd proximal phalanx
- 2nd – radial side of 3rd
- 3rd – ulnar side of 3rd
- 4th – ulnar side of 4th

➤ Nerve supply:

- Ulnar

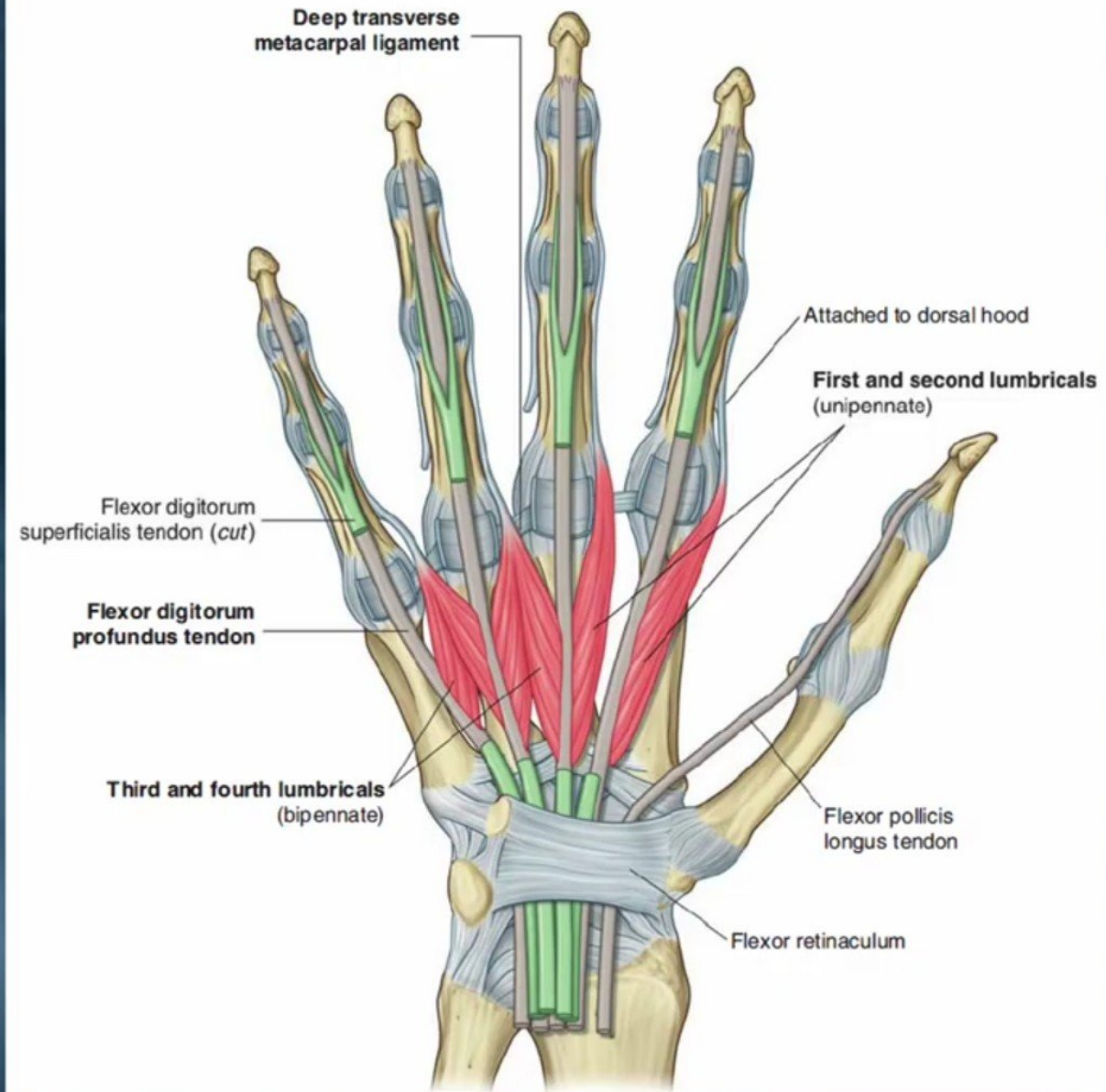
➤ Action:

- ABD of 2nd, 3rd, and 5th finger from midline



Lumbricales

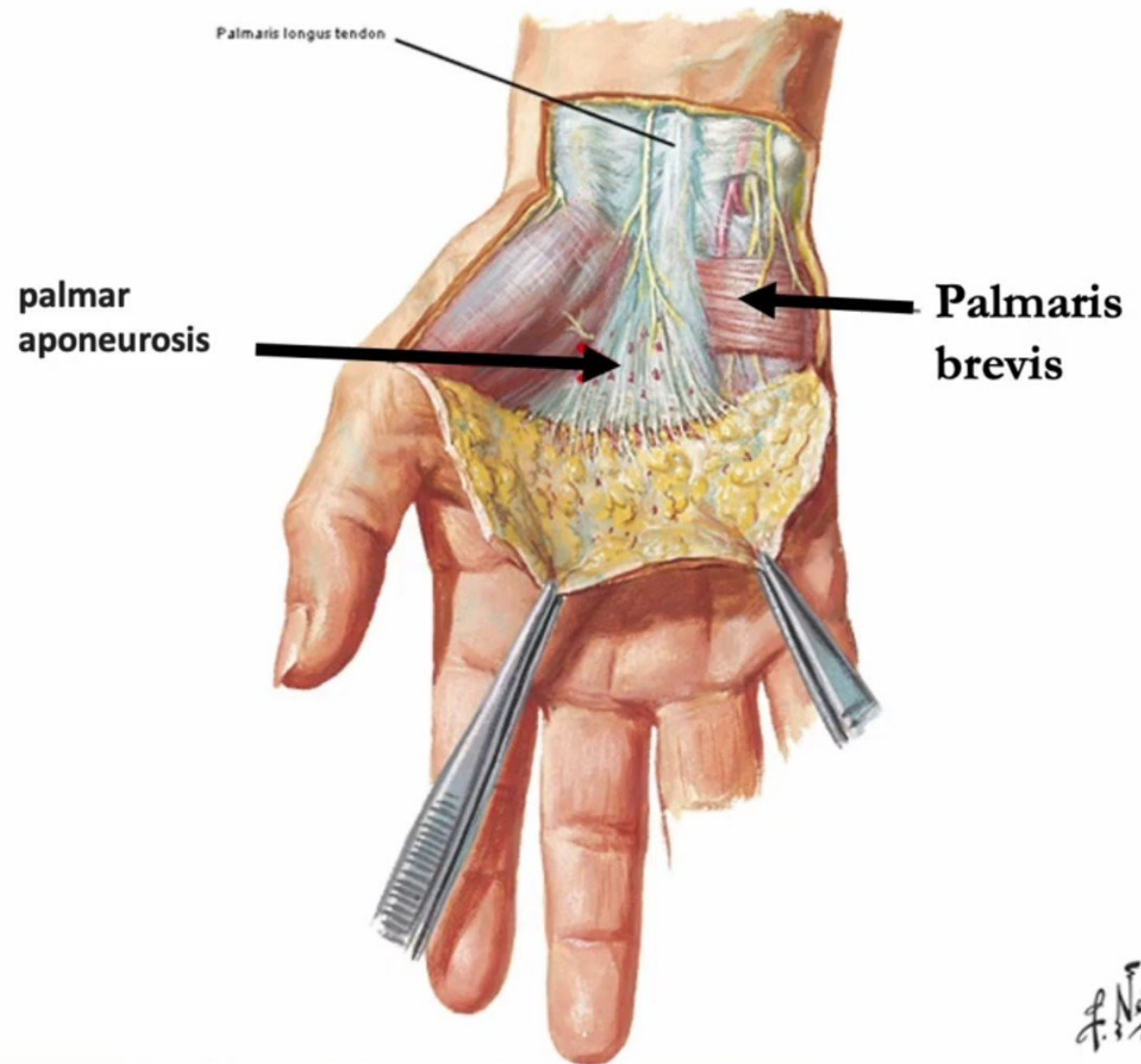
- **Origin:**
 - Tendons of FDP
- **Insertion:**
 - Extensor expansion on dorsal aspect of each digit's radial side
- **Nerve supply:**
 - 1 and 2 – median
 - 3 and 4 – ulnar
- **Action:**
 - MCP flexion 2-5 digits
 - DIP & PIP ext 2-5 digits



Palmaris Brevis

- **Origin:**
 - Flexor retinaculum
- **Insertion:**
 - Palmar surface skin on ulnar side of hand
- **Nerve supply:**
 - Ulnar
- **Action:**
 - Wrinkles skin of hand on ulnar side

Wrist and Hand: Superficial Palmar Dissection



Movements of the thumb



Abduction



Adduction



Extension



Flexion



Opposition



Reposition

Will be continue...